December 1st, 2025

Computer Vision

BSc AIDAMS ESSEC/CentraleSupélec
IDSI51006 - Winter 2025
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How to contact me ?

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Schedule

1. **November 4th** : 9am - 12pm
2. **November 4th** : 1pm - 4pm
3. **November 14th** : 9am - 12pm
4. **December 1st** : 9am - 12pm
5. **December 1st** : 1pm - 4pm
6. **December 8th** : 9am - 12pm
7. **December 8th** : 1pm - 4pm

Schedule

1. ~~**November 4th** : 9am - 12pm~~
2. ~~**November 4th** : 1pm - 4pm~~
3. ~~**November 14th** : 9am - 12pm → **10 minute quiz**~~
4. **December 1st** : 9am - 12pm
5. **December 1st** : 1pm - 4pm
6. **December 8th** : 9am - 12pm
7. **December 8th** : 1pm - 4pm → **10 minute quiz**

No final exam, instead you will complete a project and a report.

Outline for today

01. Recap of last session
02. More on CNNs
03. Exploration on Kaggle test competition
04. More deep learning for images
05. Kaggle competition modalities
06. Start of the project

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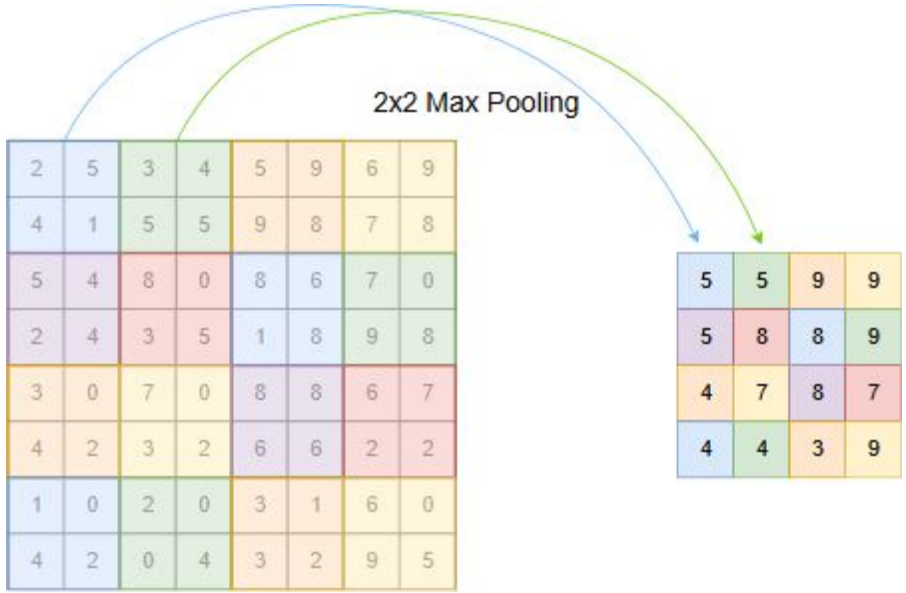
Time for the recap !

https://www.menti.com/ale3trppof5v



Last session : Pooling

- Convolution filters detect local patterns.
- Pooling compresses the information while keeping the most important activations.
- This makes the network more robust and reduces computation.
- Almost all classical CNN chain convolutional layers with pooling layers !



Source : https://hannibunny.github.io/mlbook/neuralnetworks/03ConvolutionNeuralNetworks.html

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One of the earliest CNNs : LeNet (1 to 5), 1980s

- Yann LeCun et al. developed a series of CNNs, with the task of recognising handwritten zip codes and, more generally, handwritten digits (the MNIST dataset from previous classes !).
- They consist of a sequence of Convolution and Pooling layers, as well as a Multilayer Perceptron for the output layers.

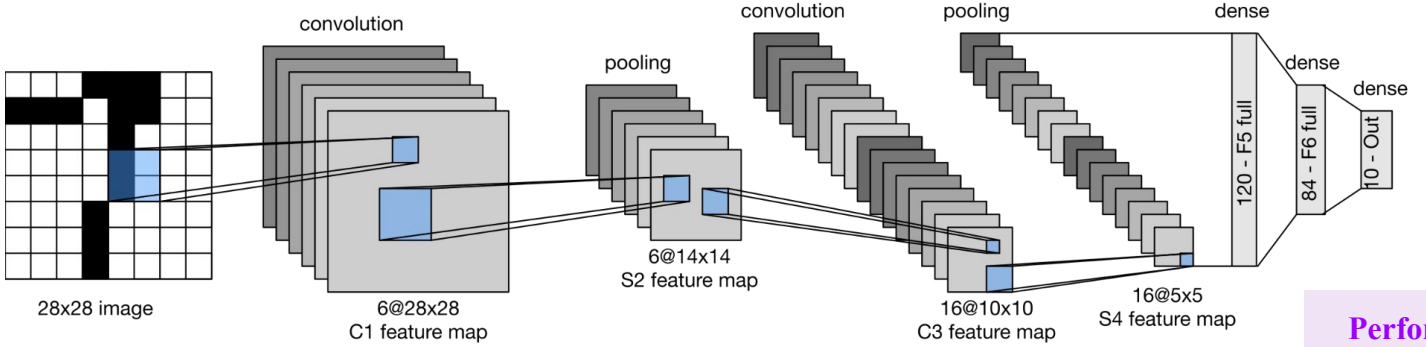


Image Source : Zhang, Aston and Lipton, Zachary C. and Li, Mu and Smola, Alexander J. - https://github.com/d2l-ai/d2l-en
LeNet (original paper) : LeCun, Y. et al. (1989). "Backpropagation Applied to Handwritten Zip Code Recognition".
LetNet-5 (most famous paper) : Lecun, et al. (1998). "Gradient-based learning applied to document recognition"
Net-1 to Net-5 comparison : LeCun, Y. (1989). "Generalization and network design strategies".

Performance on MNIST dataset

98.4%

New benchmark : ImageNet

MNIST — Modified National Institute of Standards and Technology

📅 **Date :** 1998
🎯 **Task :** Classification of handwritten digits
🗂️ **Number of classes :** 10
✅ **Evaluation :** Accuracy
🖼️ **Number of images in :**

- **Training set :** 60 000
- **Test set :** 10 000

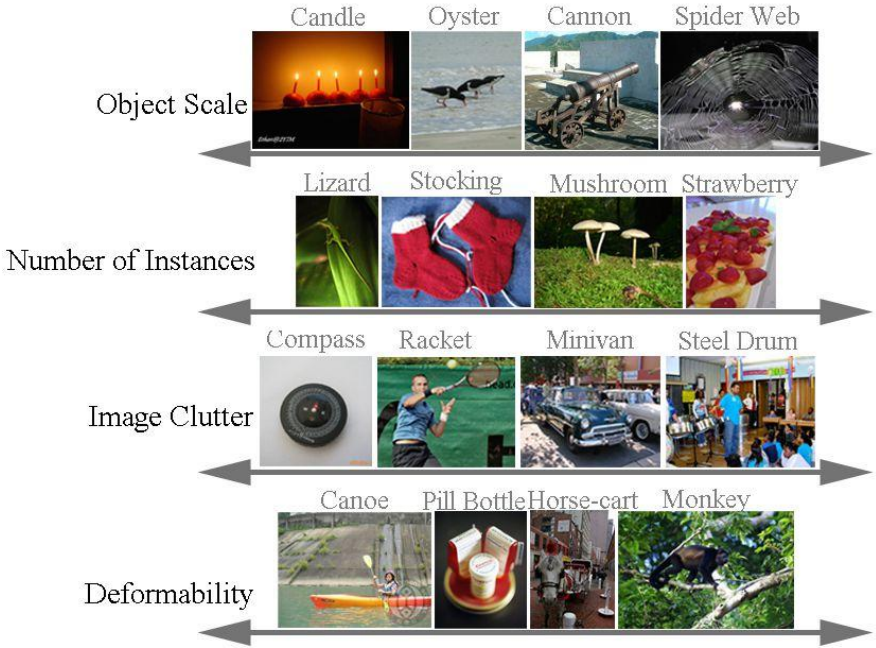
ILSVRC — ImageNet Large Scale Visual Recognition Challenge

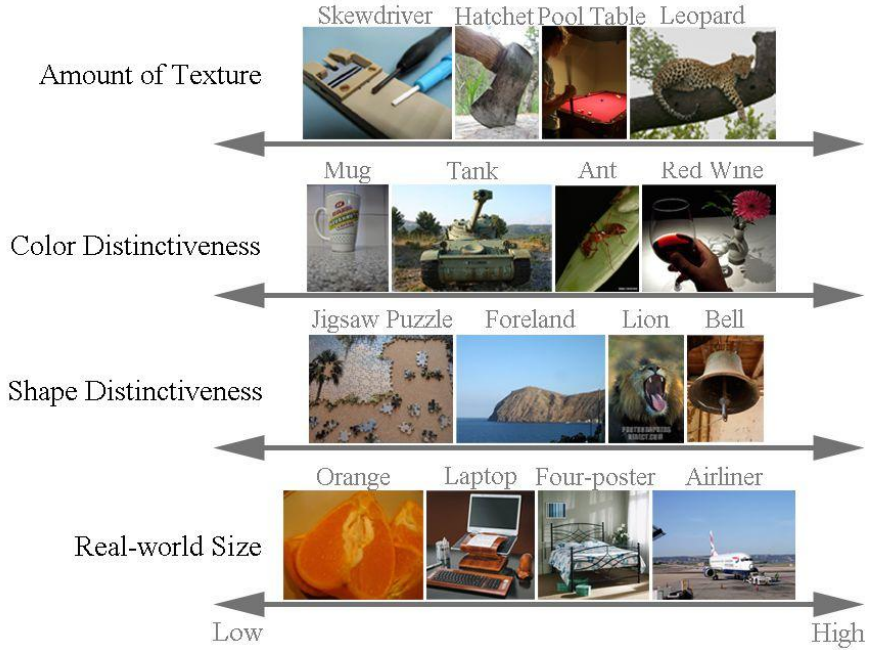
📅 **Date** : 2012-2017 (benchmark dataset)
🎯 **Task** : Classification of images
🗂️ **Number of classes** : 1 000
✅ **Evaluation** : Top-1 error, Top-5 error
🖼️ **Number of images in :**

- **Training set** : ~1 280 000
- **Validation set** : 50 000
- **Test set** : 100 000

Source : Olga Russakovsky, Jia Deng, et al. ImageNet Large Scale Visual Recognition Challenge. IJCV, 2015

New benchmark : ImageNet





Source : Olga Russakovsky, Jia Deng, et al. ImageNet Large Scale Visual Recognition Challenge. IJCV, 2015

Top-k error metric

- When classifying over 1000 classes using the accuracy might not be as informative.
- Instead, it is possible to see if the model managed to find the right target among the top-k predictions it made.
- In the case of the top-5 error, it checks if the target label is one of the top 5 predictions of the model.
- Instead of focusing on the good classifications (accuracy), it is possible to focus on the errors. That's how you obtain the top-k error metric :

$$\text{top-1 error} = 1 - \text{accuracy}$$

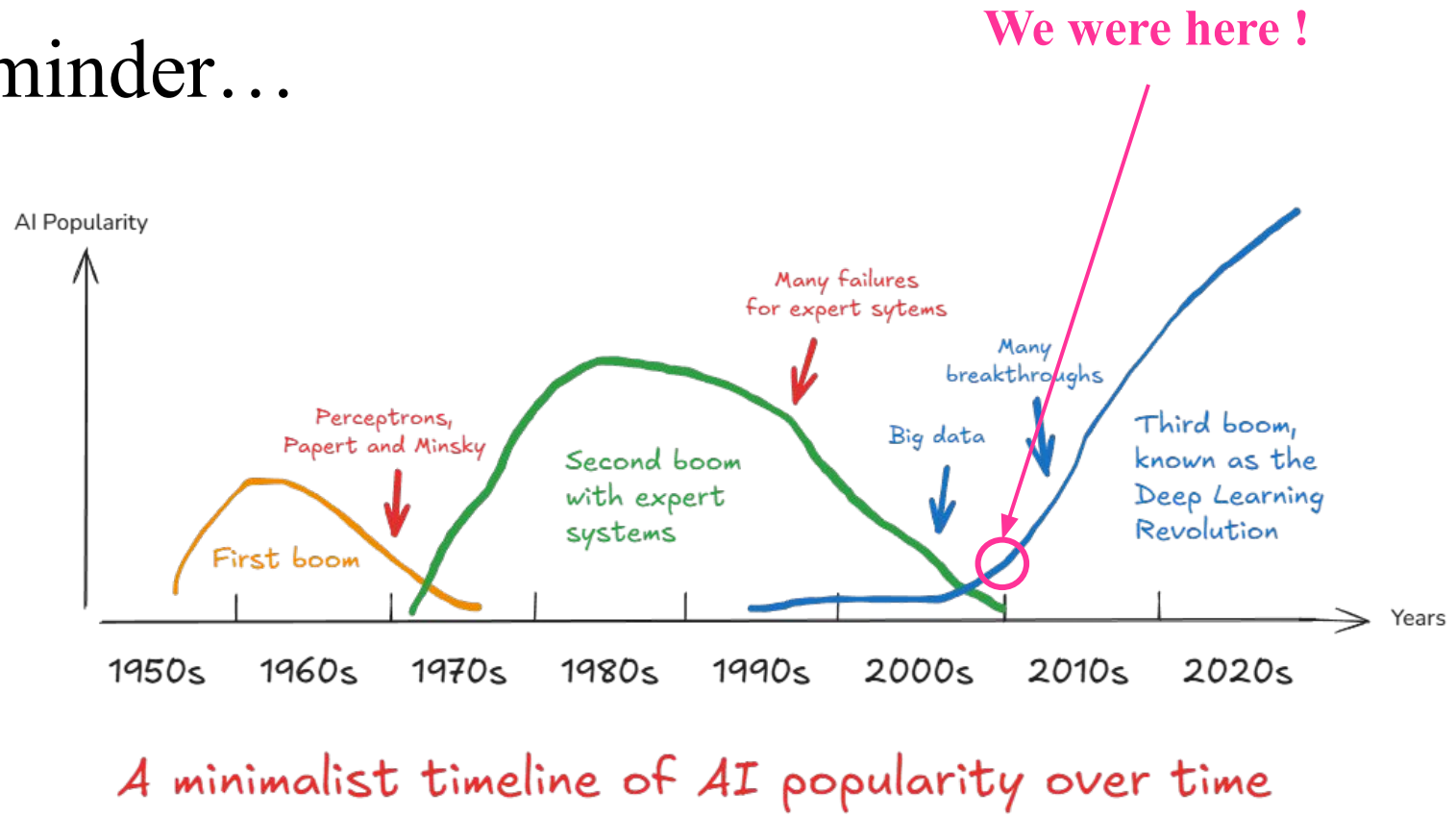
Source : Olga Russakovsky, Jia Deng, et al. ImageNet Large Scale Visual Recognition Challenge. IJCV, 2015

Until 2012, the best classification models on ImageNet were using manually crafted features (like in our first lecture !).

But why, since we knew from the 80s that CNN worked great for the MNIST dataset ?

The simple answer is : Not enough computational power !

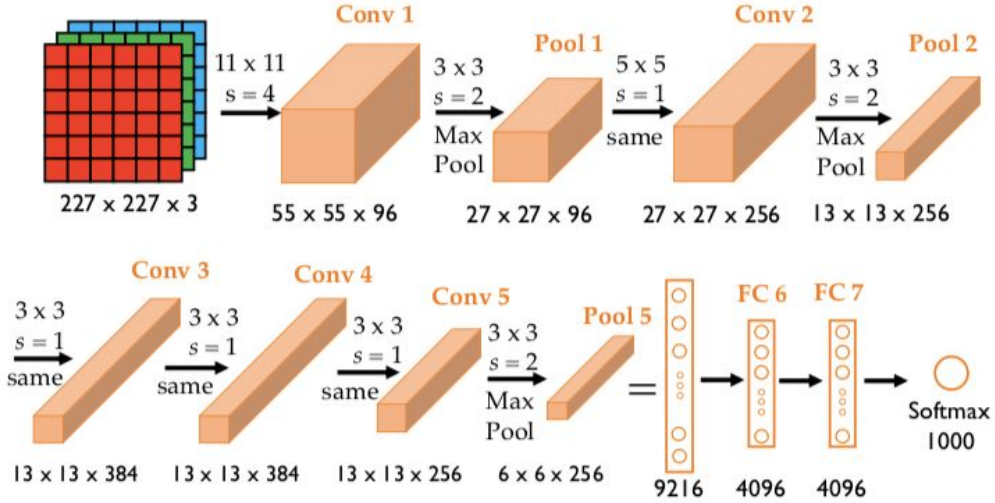
Reminder…



Source : Me (Excalidraw)

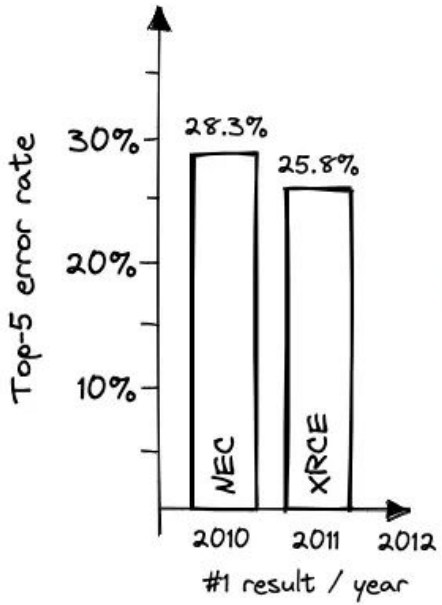
The real beginning of deep learning for images

- AlexNet ! (2012)



Source : Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. NIPS.

The real beginning of deep learning for images



Best models for the ImageNet Large Scale Visual Recognition Challenges of 2010, 2011 and 2012

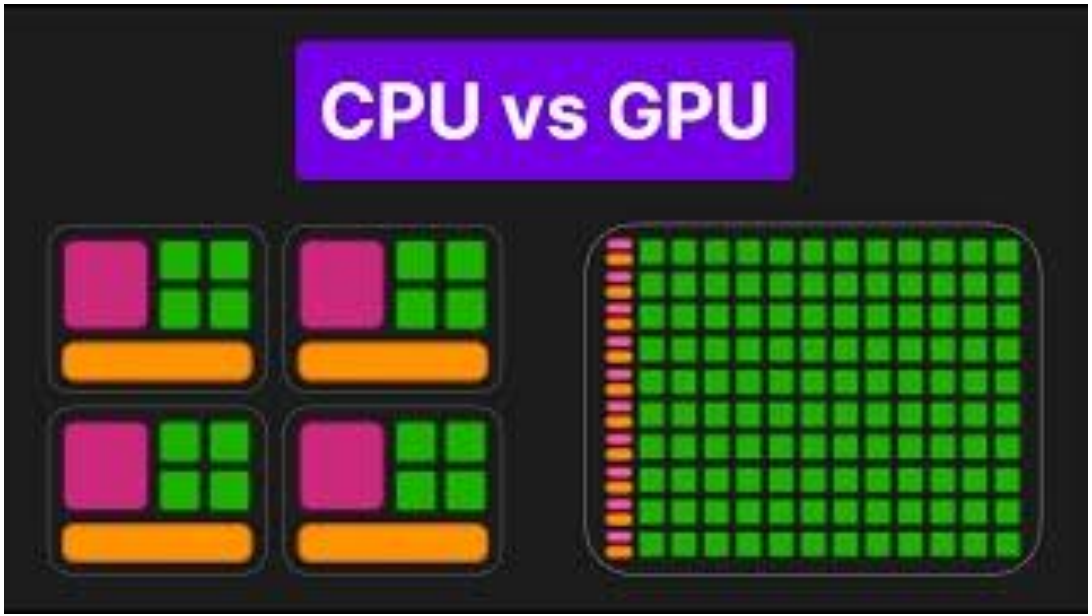
Source : James Briggs, Laura Carnevali. Embedding Methods for Image Search, Pinecone.

Why AlexNet worked better

1. It is much bigger thanks to the use of two GTX 580 **GPU**.
2. It uses ReLU activation, which diminishes **vanishing gradients** effect.
3. It adds **dropout** to avoid overfitting.
4. It increases the original dataset thanks to **data augmentation**.

Why AlexNet worked better

1. It is much bigger thanks to the use of two GTX 580 **GPU**.

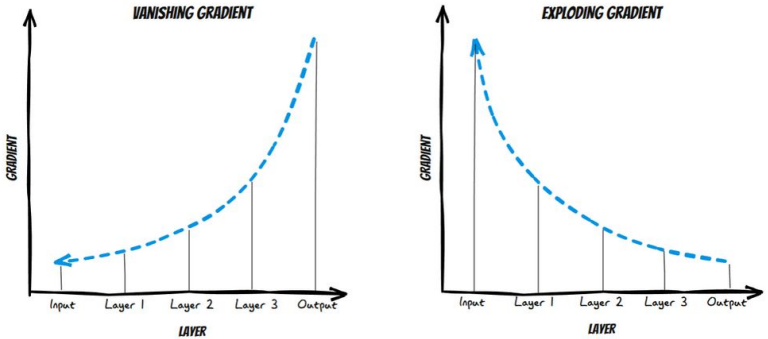


Link : https://www.youtube.com/watch?v=Axd50ew4pco

Why AlexNet worked better

2. It uses ReLU activation, which diminishes **vanishing gradients** effect.

- Remember the chain rule to compute backpropagation ?
- As the number of forward propagation steps increases…
- … the gradients of the first weights have more and more multiplications !
- This can exponentially increase or decrease the gradient magnitude !

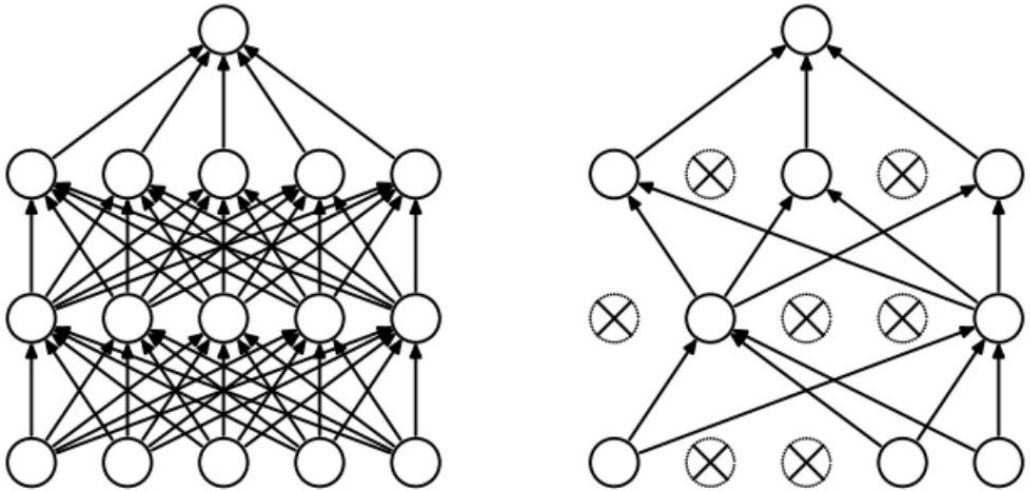


Vanishing gradient effect is the most common. It gets too difficult to **update the first layers** as the gradient magnitudes become **too small**.

Source of gif image : https://medium.com/dscier/how-to-deal-with-vanishing-and-exploding-gradients-in-neural-networks-24eb00c80e84

Why AlexNet worked better

3. It adds **dropout** to avoid overfitting.



(a) Standard Neural Net

(b) After applying dropout.

Srivastava, Nitish, et al. ”Dropout: a simple way to prevent neural networks from overfitting”, JMLR 2014.

Why AlexNet worked better

4. It increases the original dataset thanks to **data augmentation**.

❓ **What's data augmentation ?** It's a perturbation of existing data to enrich the training dataset and avoid overfitting.

🧪 Let's see examples of **geometric** and **color space** transformation, as well as **noise injection**.



Geometric transformations

Cropping





Geometric transformations

Affine transformations : Rotation





Geometric transformations

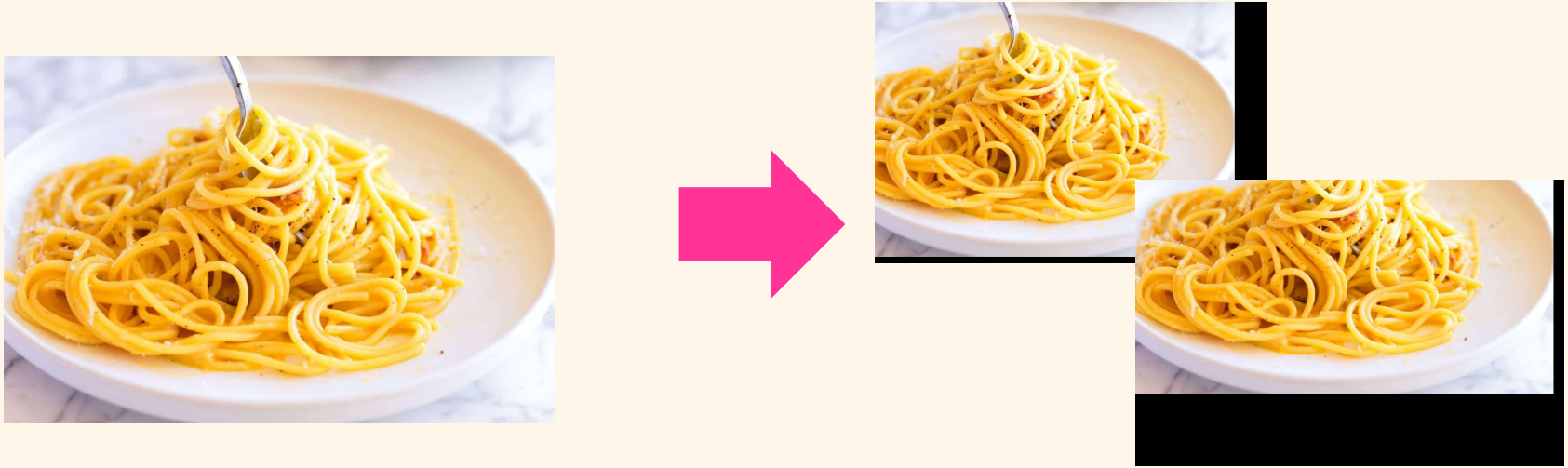
Affine transformations : Reflection





Geometric transformations

Affine transformations : Translation



Geometric transformations

Elastic distortion





Geometric transformations

Morphing







Color space transformations

Brightness adjustment





Color space transformations

Contrast adjustment





Color space transformations

Saturation adjustment





Color space transformations

Noise injection : Gaussian Noise





Color space transformations

Noise injection : Salt and Pepper





Out of 1 image, we can create 20 more !

Feedback on quiz 1

Everybody did great !

Expectations for question 1 :

- Understanding that we are working with matrices or tensors.
- Grayscale = 1 matrix
- Coloured = 3 or 4 matrices (RGB/BGR, RGBA)
- Each value in the matrices represents the intensity of a the colour of the channel.
- 200x200 = 40 000 pixels (x 3 = 120 000 so 120 000 different values)

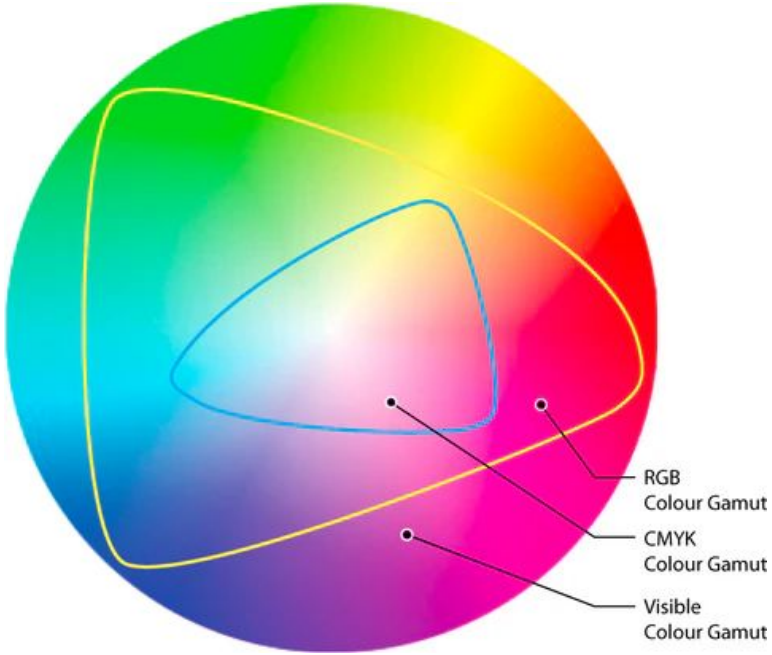
Feedback on quiz 1

Expectations for question 2 :

- Suggesting a method seen in class : Edge/corner detection, thresholding, extracting mean value per channel…
- The method should not be deep learning related !
- Explanation in one or two sentences.

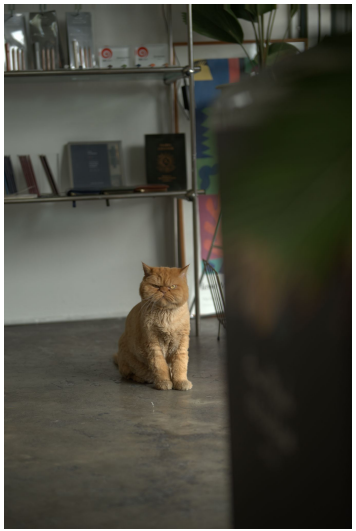
More on images

- I presented to you a way to store images using **8-bit** values (from 0 to 255), which is widely used in many applications.
- But it is possible to **decrease** or **extend** this storage to capture a different range of colours and intensities !
- Indeed, 8-bit RGB does not reproduce all the colours a human eye can see.

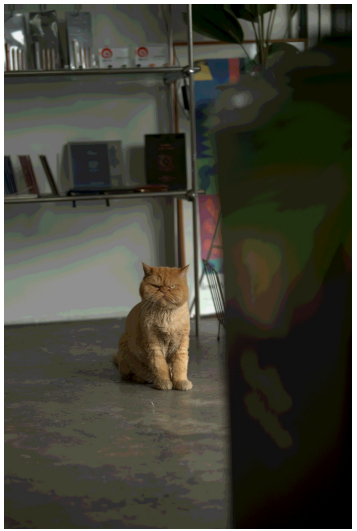


More on images

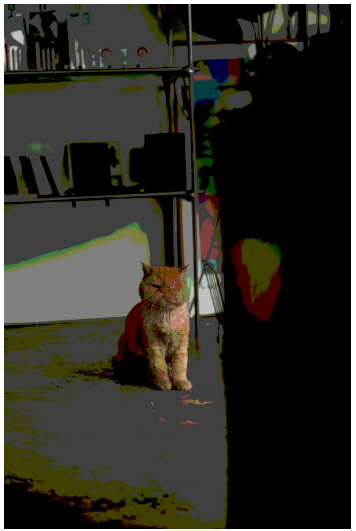
Examples of storage increase : RAW files (here ARW = 14 bytes)



8-bit



4-bit



2-bit

Source of RAW file : https://www.signatureedits.com/free-raw-photos/

More on images

Examples of storage increase : HDR10 (SDR = 8-bit)



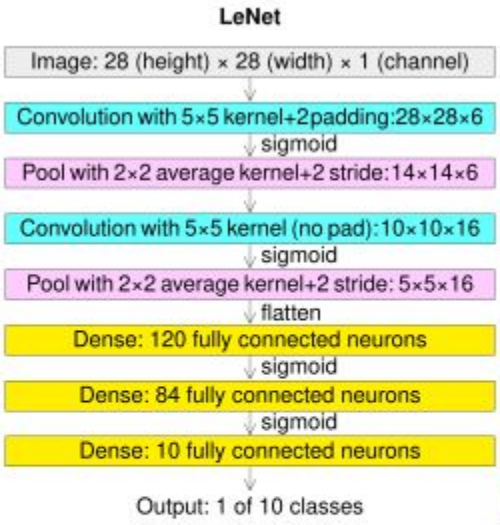
Source of image : https://blog.eavs-groupe.com/guides-et-dossiers/dossier-quest-ce-que-le-hdr-en-quoi-est-il-important/

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Exploration on Kaggle test competition

Goal : Reimplement LeNet, with the following architecture, on the partial MNIST dataset given in the Kaggle competition. Submit in the right format.



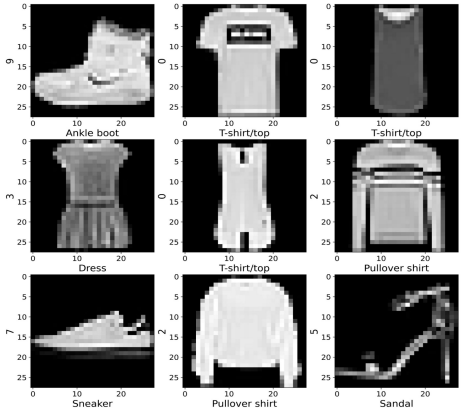
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Now that we have good tools… which tasks can we do ?

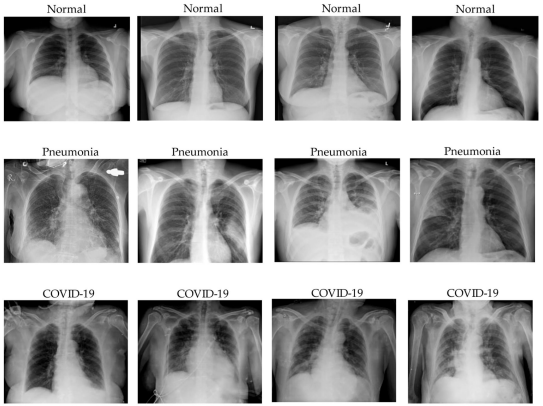
Firstly, working with images…

Classification



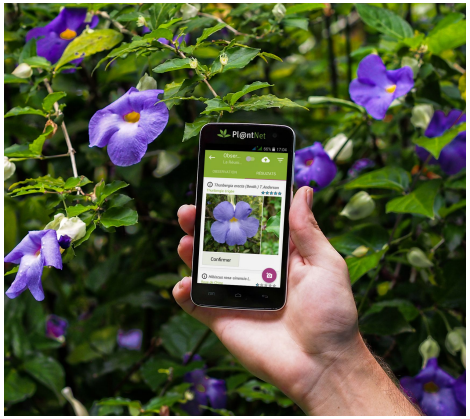
Fashion & E-commerce

Classify images from sellers' listings to improve search feature, or to quickly remove unauthorised listings.



Medical imaging

X-ray, Histopathology or Skin Lesion classification

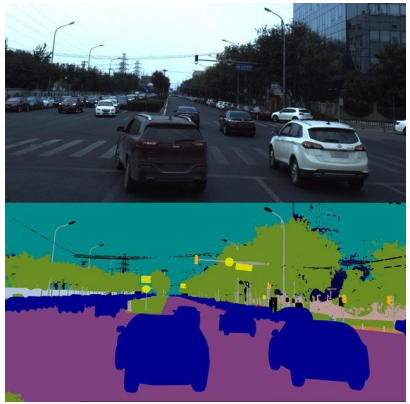


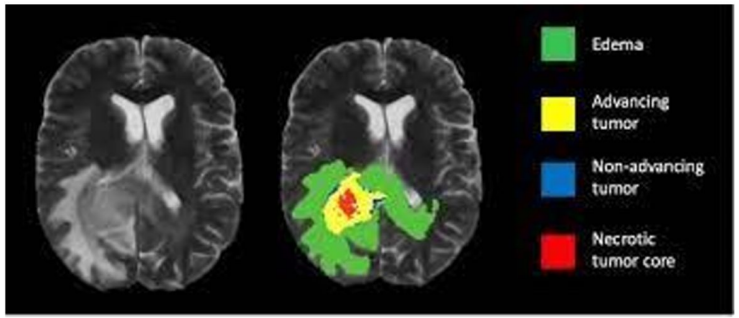
Agriculture & Environment

Plant species or disease identification, fruit maturity classification

Sources : Fashion MNIST (Zalando), COVID-19 Chest X-ray Classification (Le Dinh et al.), PlantNet

Segmentation







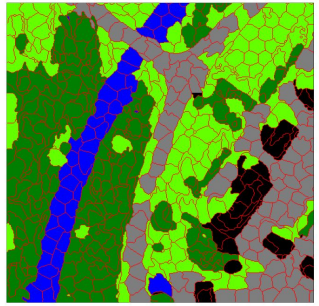


Fig 1. Example of Image Segmentation in medical imaging

Autonomous driving
Road scene parsing: road, sidewalk, car, sky, trees...

Medical imaging
Tumor boundary segmentation on MRI, lesion segmentation in dermatology (melanoma borders)

Satellite images
Land-cover segmentation (urban, vegetation, water, bare soil), road segmentation.

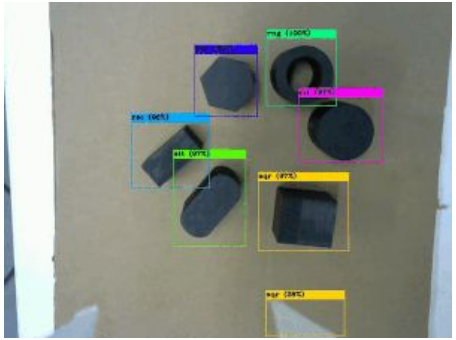
Sources : Muhammad Khan et al., Daniel Roshan., Längkvist Martin et al.

Object detection





(**a**) Attended object (**b**) Unattended object



Autonomous driving
Next step, with more precise bounding boxes to detect pedestrians, cars…

Security & Surveillance
Detecting people in restricted areas, weapons or dangerous objects.

Robotics
Object tracking for grasping, manipulation, detecting obstacles for navigation.

Sources : Elallid Badr et al., Luna Elena et al., https://github.com/Walid-khaled/

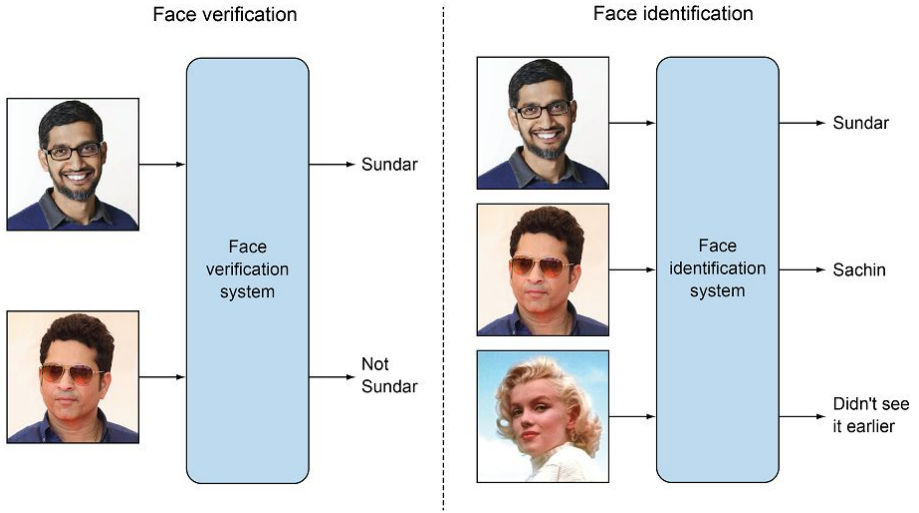
Face verification and identification

Face verification :

- Phone authentication
- Border control

Face identification :

- Photo organisation
- Surveillance systems (not allowed, but experimentations without face recognition during the 2024 Olympics)
- Law enforcement (not allowed in real time, but allowed a posteriori)

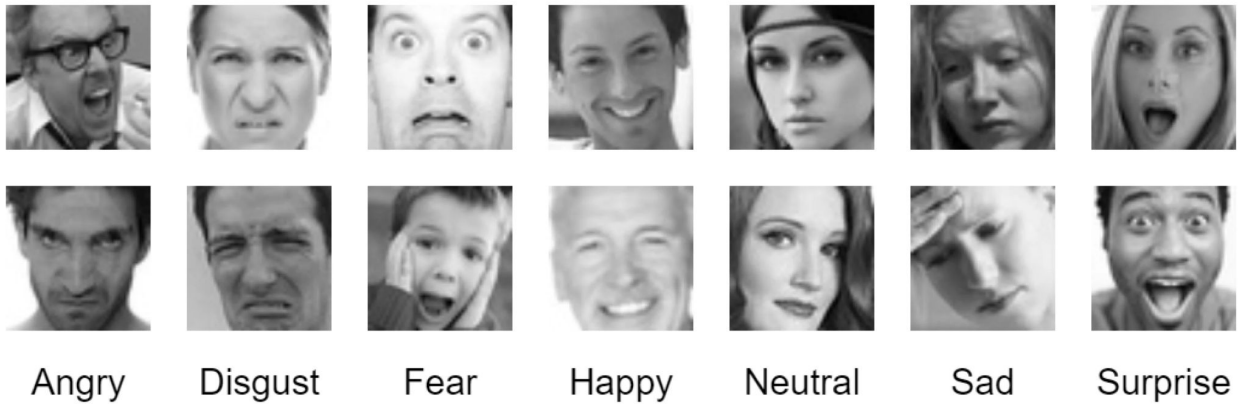


Sources : Deep Learning for Vision Systems, Elgendy, Mohamed

Emotion recognition

Applications :

- Human-Robot interactions
- Lying detector for job interviews (now forbidden by the AI Act !)
- Evaluate emotional responses to ads (now forbidden by the AI Act !)



Sources : Białek et al.

Most tasks can be adapted to videos…

Depth estimation



Sources : https://www.youtube.com/watch?v=paHCQDb3eNw

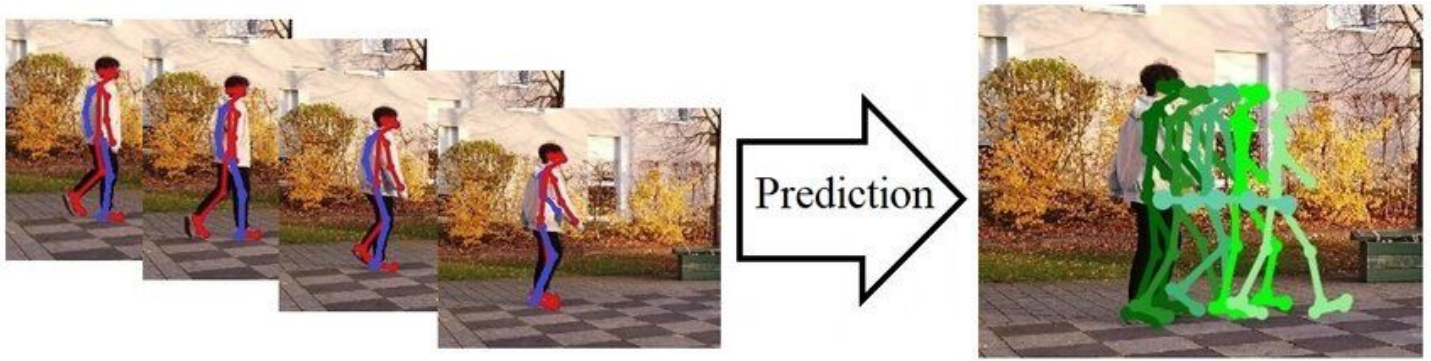
Human pose estimation



Sources : https://www.youtube.com/watch?v=pW6nZXeWlGM

Some tasks are specific to videos

Motion prediction



Sources : Mascaro Esteve et al.

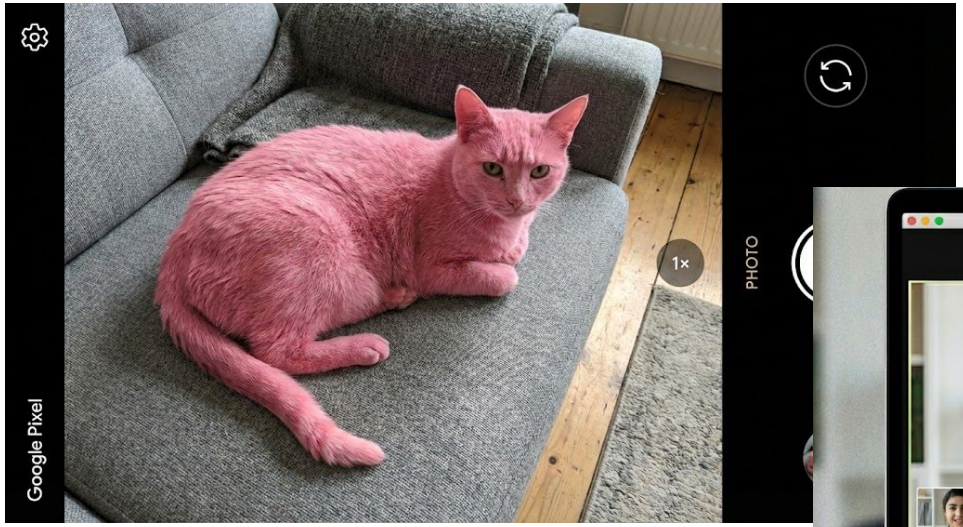
And, finally, some tasks do not generate labels or landmarks, they generate images !

Style transfer



Sources : Chen Jiyou et al.

Text-based Image generation





Sources : Nano Banana Pro

This is just a broad overview…

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Goal : Classify emojis based on their style


Apple


Google


Facebook


Samsung


Whatsapp


Messenger


Mozilla

- You will team up by 3 or 4 to analyse the given dataset.
- You will write a report to document your trials and errors.
- Deadline… to be discussed !

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https://www.kaggle.com/t/e2037a5df75a409a881c56dc6abe395a

See you next week !